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GRAU DE MATEMÀTIQUES

Treball final de grau

GRAPH NEURAL NETWORKS FOR AUTONOMOUS THEOREM PROVING

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Barcelona, March 7, 2026

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Abstract

The main goal of this work is to prove the existence of.....

Notation: We will work over an algebraically closed field K of characteristic zero.

2020 Mathematics Subject Classification. 11G05, 11G10, 14G10

Chapter 1

Generalized Algebraic Theories

Our goal is to generate a term of a specific type in a given theory. The goal of this chapter is to give meaning to the concept of term, type and theory, while introducing some tools that we will use later when we present the algorithm.

By theories we mean Generalized Algebraic Theories (or GATs), introduced by John Cartmell in his thesis [1] [2], they give us both a very idiomatic way of presenting theories and powerful interpretation tools like Context Categories, introduced later in this section. The notion of generalized algebraic theory is a generalization of the notion of many-sorted algebraic theory in the following manner: while in Martin Löff's many-sorted algebraic theories sorts are constant types, interpreted as sets, in generalized algebraic theories sorts might be constant or variable, over a specified range of types, and variable types are interpreted as families of sets. So a generalized algebraic theory consists of:

1. a set of sorts, that can be constant or variable;
2. a set of operators, with the sort of the arguments and value specified;
3. a set of axioms, where each axiom is an identity between similar expressions.

Let's present category theory for example: The set of sorts contains two elements Ob and Hom. Ob is a constant type and Hom is a variable type, for a pair of terms of type Ob, name them x, y, we have a type $Hom(x, y)$. Here we see a key point in general algebraic theories, types vary over terms, no over types. The set of operators also has two elements id and $\circ$. The id operator has one argument x of type Ob, and returns a value of type $Hom(x, x)$. This shows how the type of value of an operator might vary depending on the arguments. The $\circ$ operator (or composition) expects 5 arguments, three of type Ob, name them x, y, z, one of type $Hom(x, y)$, say f, and one of type $Hom(y, z)$, named g, and it returns a value of type $Hom(x, z)$. Here we see that the type of an argument might also vary

depending on a previous argument. Informally, instead of writing $\circ(x, y, z, f, g)$ we can write $\circ(f, g)$, as the first three arguments might be inferred from the last two. This notation will be discussed in a further section.

Now we will introduce the notation we will be using when talking about terms and types. This notation will be formally defined in a later section. So consider you have a set of variables V, we will associate terms to variables as required. When asserting that a term t is of type Δ we will write $t : \Delta$, but if t has variables $x_1, \ldots, x_n$ occurring in it this assertion does not make sense unless we assign some types to $x_1, \ldots, x_m$. To do this we will write:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

that reads as: if x_1 is a variable of type Δ_1, ...and if x_n is a variable of type Δ_n, then t is a term of type Δ. Similarly, if we want to assert that if x_1 is a variable of type Δ_1, ...and if x_n is a variable of type Δ_n, then Δ is a type, we can write:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

this assertions are called rules.

Chapter 2

The Proposed Model

Consider the presentation of a theory U. The algorithm has one goal, given a derived T-rule $\dfrac{Q}{\Delta \text{ is a type}}$ of the theory U it will try to create a term of type Δ under the context Q, i.e. finding a realization from the context Q to the context (Q, Δ). To do this we will consider the context category $C(U)_Q$, to limit ourselves to reasoning under the context Q.

Let G be a finite subcategory of $C(U)_Q$ containing both Q and (Q, T). We apply iteratively the following steps

1. assign a label to all the objects of G in 3 categories: 'keep', 'expand' and 'delete'. (this is where the AI model comes in)

2. Let G_1 be the category consisting of all objects of G not labeled as 'delete'.

3. For each object X of G labeled as 'expand' apply the following expansion rules:

   (a)

TODO: intoroducir $\leq$
TODO: operad

Bibliography

[1] John Cartmell, *Generalized Algebraic Theories and Contextual Categories*, Ph.D. thesis, Oxford University, 1978.

[2] _____, *Generalised algebraic theories and contextual categories*, Annals of Pure and Applied Logic **32** (1986), 209–243.